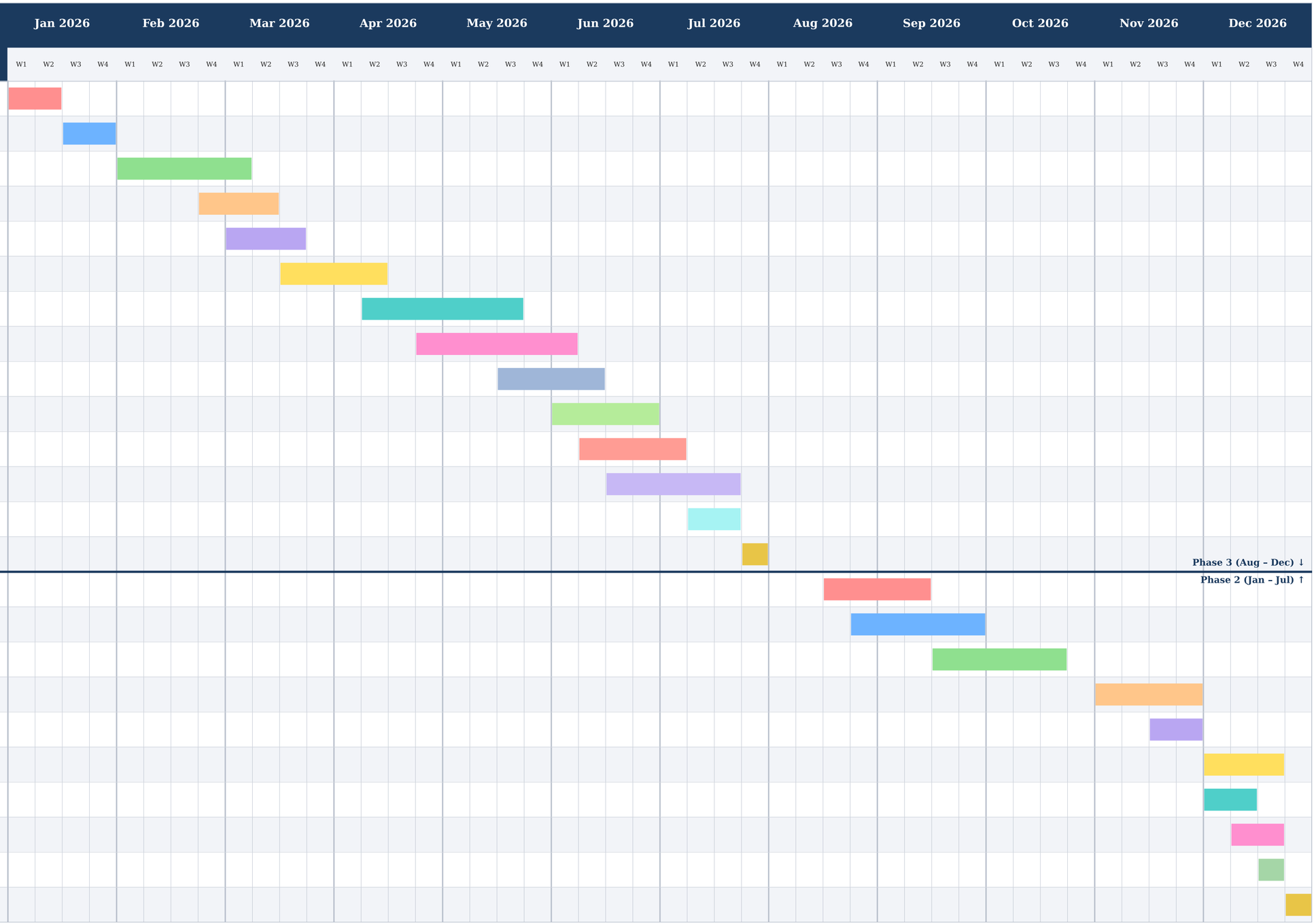
Gantt Chart – Capstone Project (Phase 2 & Phase 3)

Federated Learning For Privacy-Preserving Pneumonia Detection Using Chest X-Ray Images

No.	Tasks / Modules
1	Literature Survey & Problem Definition
2	Dataset Acquisition (RSNA)
3	Data Preprocessing (DICOM decode, MONOCHROME1 correction, normalize, resize)
4	Dataset Splitting & Non-IID Client Partitioning
5	Environment Setup & Tool Integration (PyTorch, Flower, etc.)
6	Model Architecture Implementation (DenseNet-121)
7	Federated Learning Core (SCAFFOLD) (Client & Server)
8	Training & Hyperparameter Tuning
9	Evaluation & Threshold Tuning (Accuracy, Precision, Recall, F1)
10	Streamlit Demonstration Application (Upload, Predict, PDF Report)
11	Testing & Debugging
12	Report Writing (Phase 2)
13	PPT Preparation
14	Review 2 Submission & Final Demonstration
	Phase 3 (Aug – Dec) ↓
	Phase 2 (Jan – Jul) ↑
15	FedAvg Comparison Run (same 4 clients, no SCAFFOLD)
16	Additional Data Source Integration (extra pediatric / adult dataset)
17	Adult Accuracy Improvement (weighted aggregation, more rounds)
18	Grad-CAM Explainability – Novelty Progress (heatmaps showing where the model looks)
19	Retraining & Re-evaluation (Accuracy, Precision, Recall, F1)
20	Privacy Layer – Progress (differential privacy on client updates)
21	Paper Revision (Grad-CAM, privacy, FedAvg results)
22	Testing & Debugging (app and training pipeline)
23	Final Report & PPT Preparation
24	Final Submission & Demonstration



Phase 2: Jan–Jul 2026 | Phase 3: Aug–Dec 2026 (planned dates, adjustable)